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Beyond the Gutenberg-Richter law? Testing Dragon-king distributions in seismicity of China

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Motivation

Whether large earthquakes enjoy some distinguishable properties that could make them predictable? This question is motivated both by the quest of understanding the physics of earthquakes and their spatio-temporal organization and by the profound implications this would have for seismic risk assessment and mitigation.

Dragon-king events and dragon-king theory

Dragon-kings (DK) embody a dual metaphor, expressing that an event is both exceptionally large ("king") and originates from unique circumstances ("dragon") compared to its peers. The hypothesis proposed by Sornette (2009), with a premonitory discussion emphasizing the predictability of catastrophes that are outliers (Sornette, 1999), posits that DK events arise from distinct mechanisms that occasionally amplify extreme events, resulting in the creation of runaway disasters on the downside as well as extraordinary opportunities on the upside. Possible generative mechanisms, statistical characteristics, modeling predictions, and risk management strategies of dragon-king events have been explored in (de S. Cavalcante et al., 2013; Sornette, 2009; Sornette and Ouillon, 2012).

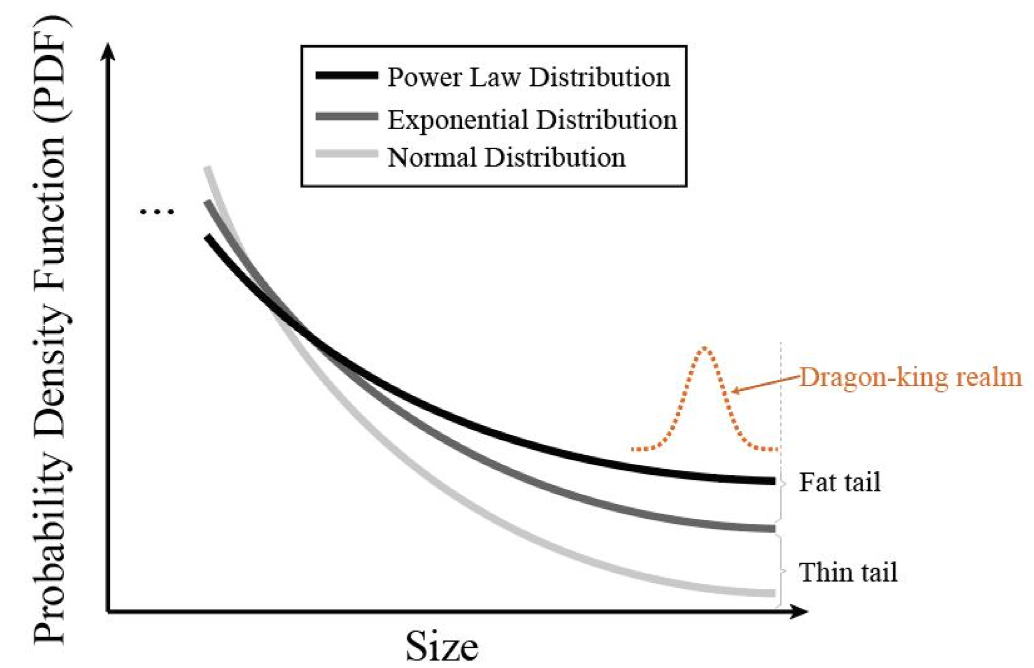

Fig. 1 A schematic diagram of the observations of dragon-king events appearing as outliers in the tail of a power law distribution. Although dragon-king events are extreme, their occurrence probability is much higher compared to Gaussian and exponential distributions, and even significantly higher than the power-law distribution. Typically, if the probability density function decays faster than an exponential distribution in the tail, it is referred to as a thin tail, such as for a Gaussian distribution; whereas a decay slower than an exponential is known as a fat tail (sub-exponential in mathematical terms). A specific example of a fat tail distribution is the power law distribution. The term "heavy tail" is technically used to refer to an asymptotic power law decay with exponent smaller than 2, so that the variance of the distribution is mathematically infinite. The dragon-king regime can express itself as a shoulder or as a change of speed of decay in the far-right end of the distribution.

Framework for detecting dragon-king earthquake

The framework we developed for detecting dragon-king earthquakes comprises four main steps, each carefully designed to address this critical challenge.

1. Defining the Study Region: The first step focuses on identifying clusters of seismicity using kernel density estimation (KDE). This technique ensures that the study domain reflects natural groupings of earthquakes rather than arbitrary boundaries. For further analysis, this step is pivotal for identifying a set of events that are likely to be associated with the same seismogenesis. Studying the Gutenberg-Richter distribution within coherent seismicity regions, rather than in aggregate, is crucial to uncover localized geophysical variations and distinct earthquake types like dragon-kings, which broad-scale analysis obscures.

2. Identifying Dragon-King Candidates: Once the study regions are defined, potential dragon-king earthquake candidates are identified by analyzing the statistical distribution of seismicity in each region separately. This involves examining the tail behavior of the FMD in each region and searching for anomalies that deviate markedly from the expected patterns of the generating process.

3. Rigorous Statistical Testing: The next step applies rigorous statistical methods to individually test whether the identified candidates truly represent dragon-king earthquakes. These tests are specifically designed to ensure that the observed anomalies are not due to random variations or errors in data processing but are statistically significant deviations indicative of unique underlying processes.

4. Validation of Dragon-King Events: If dragon-king earthquakes are detected, the final step involves validating these events through further statistical analysis using block tests. This may also include analyzing the geological and geophysical context and physical mechanisms associated with the events, and comparing them to similar phenomena in other regions or contexts.

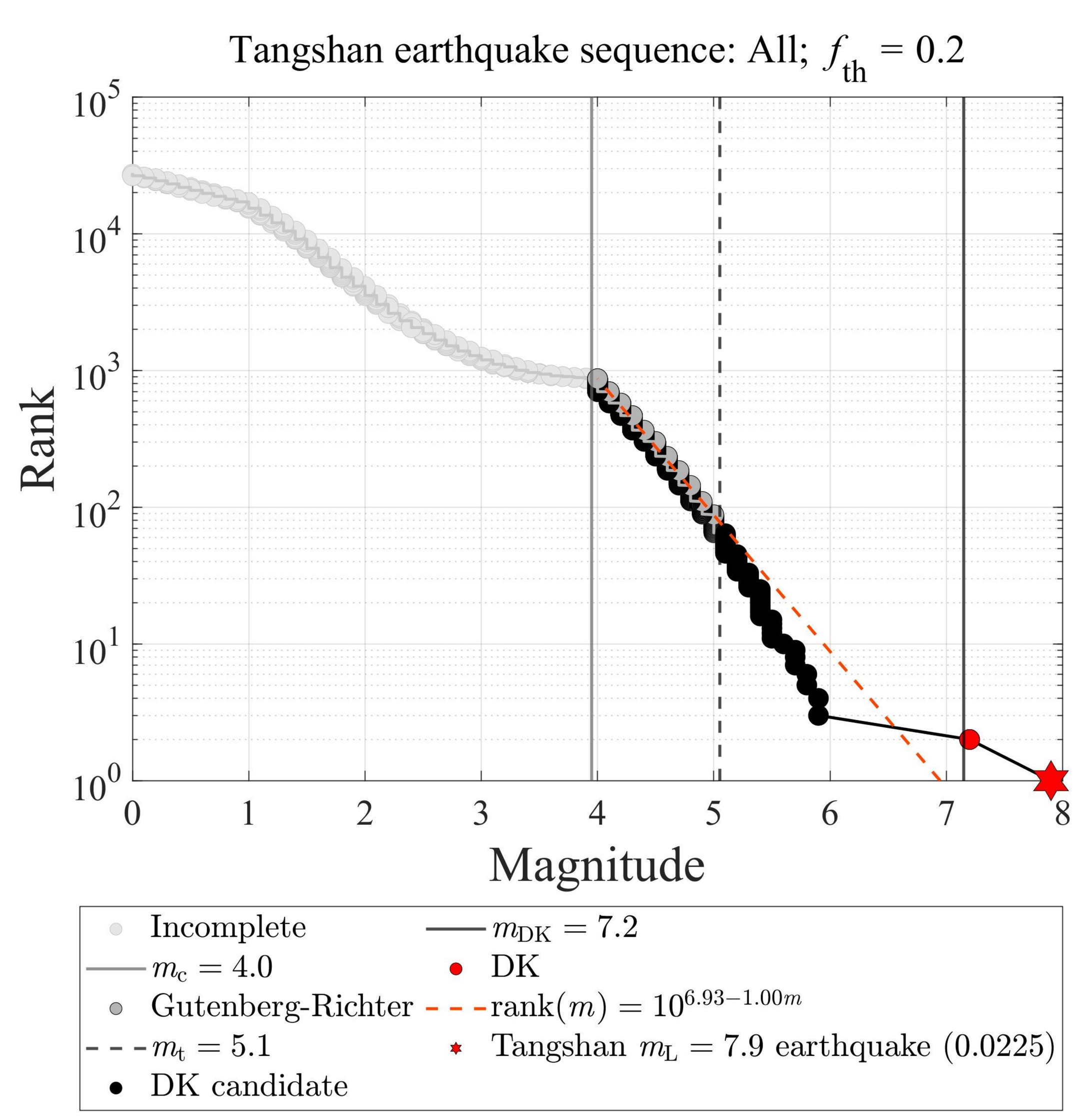

Fig. 2 Magnitude rank-ordering plots (non-normalized complementary cumulative frequency-magnitude distribution, CCFMD) for combined pre-mainshock and post-mainshock sequences of the m_L 7.9 Tangshan earthquake on July 28, 1976, for density threshold f_{th} = 0.2. The dragon-king earthquake testing framework developed in this study divides the frequency-magnitude distribution into three distinct regions based on the completeness magnitude m_c (vertical gray line) and the dragon-king earthquake candidate transition magnitude m_t (vertical gray dashed line): the incomplete part (light gray solid circles), the Gutenberg-Richter part (dark gray solid circles), and the dragon-king candidate part (black solid circles). If dragon-king earthquakes (red solid circles) are identified with a significance level α = 0.05, they are bounded by the minimum magnitude among them, referred to as the dragon-king transition magnitude m_{DK} (vertical black line). The CCFMD as a function of magnitude, CCFMD(m) = 10^{a-bm}, representing the Gutenberg-Richter law, is fitted to the data above m_c and below m_{DK}, and is shown by the red dashed line. The p-value of the m_L 7.9 Tangshan earthquake as a potential dragon-king earthquake, obtained with the max-robust-sum (MRS) test statistic using the other events in the sequence, is displayed in the parentheses of the legend.

Are Haicheng and Tangshan Earthquakes Dragon-Kings?

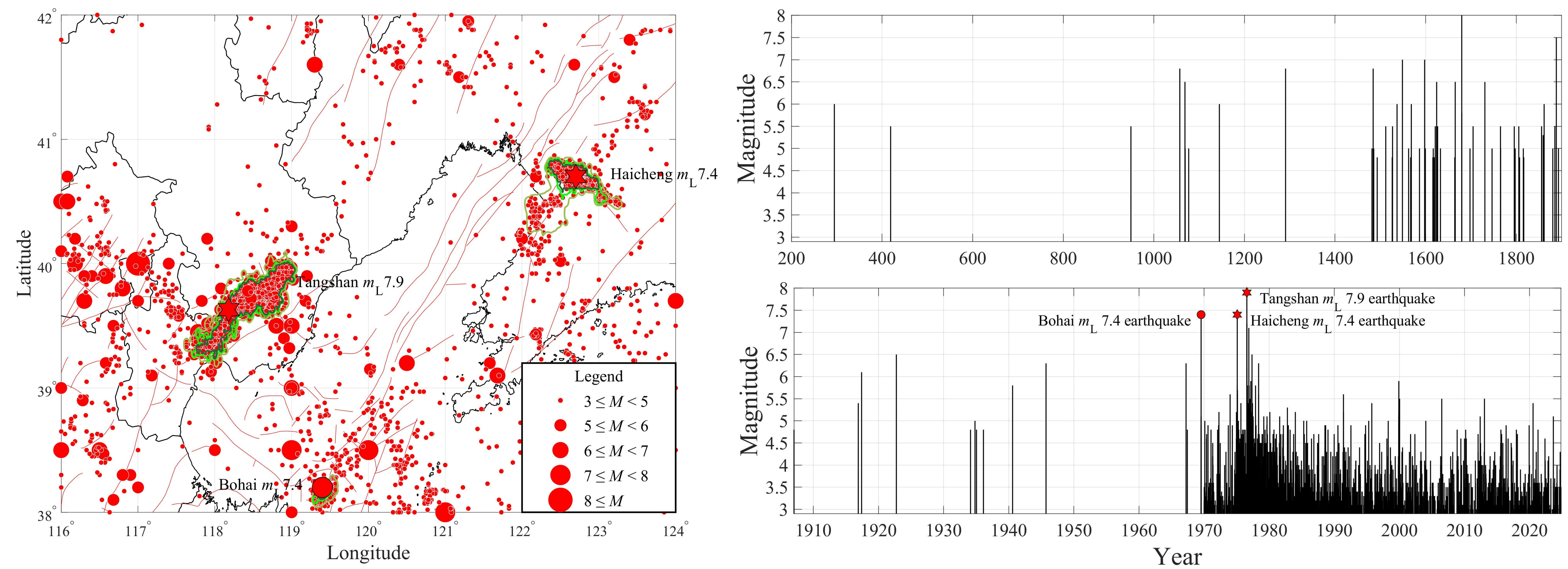

Fig. 3 Epicenters of earthquakes with $m_L \ge 3.0$ that occurred from 200 to 2024 in the vicinity of Bohai Bay, China from the China Earthquake Networks Center (CENC). Thin black lines indicate the provincial administrative boundaries of China, while thin red solid lines indicate known mapped faults. The red hexagonal stars mark the m_L 7.4 Haicheng earthquake on February 4, 1975, and the m_L 7.9 Tangshan earthquake on July 28, 1976. The red solid circle marks the m_L 7.4 Bohai earthquake on July 18, 1969, which generated a prominent aftershock cluster. Contours delineate the spatial extent of clusters that encompass the three mainshocks at density thresholds f_{th} =0.05 (light green), f_{th} =0.15 (medium green), and f_{th} =0.25 (dark green). (Right top) The magnitude-time plot of 60 earthquakes with intensity magnitude $m_I \ge 4.8$ that occurred from 200 to 1900 and (Right bottom) the magnitude-time plot of approximately 3,900 earthquakes with $m_L \ge 3.0$ that occurred from 1900 to 2024 in the vicinity of Bohai Bay, China.

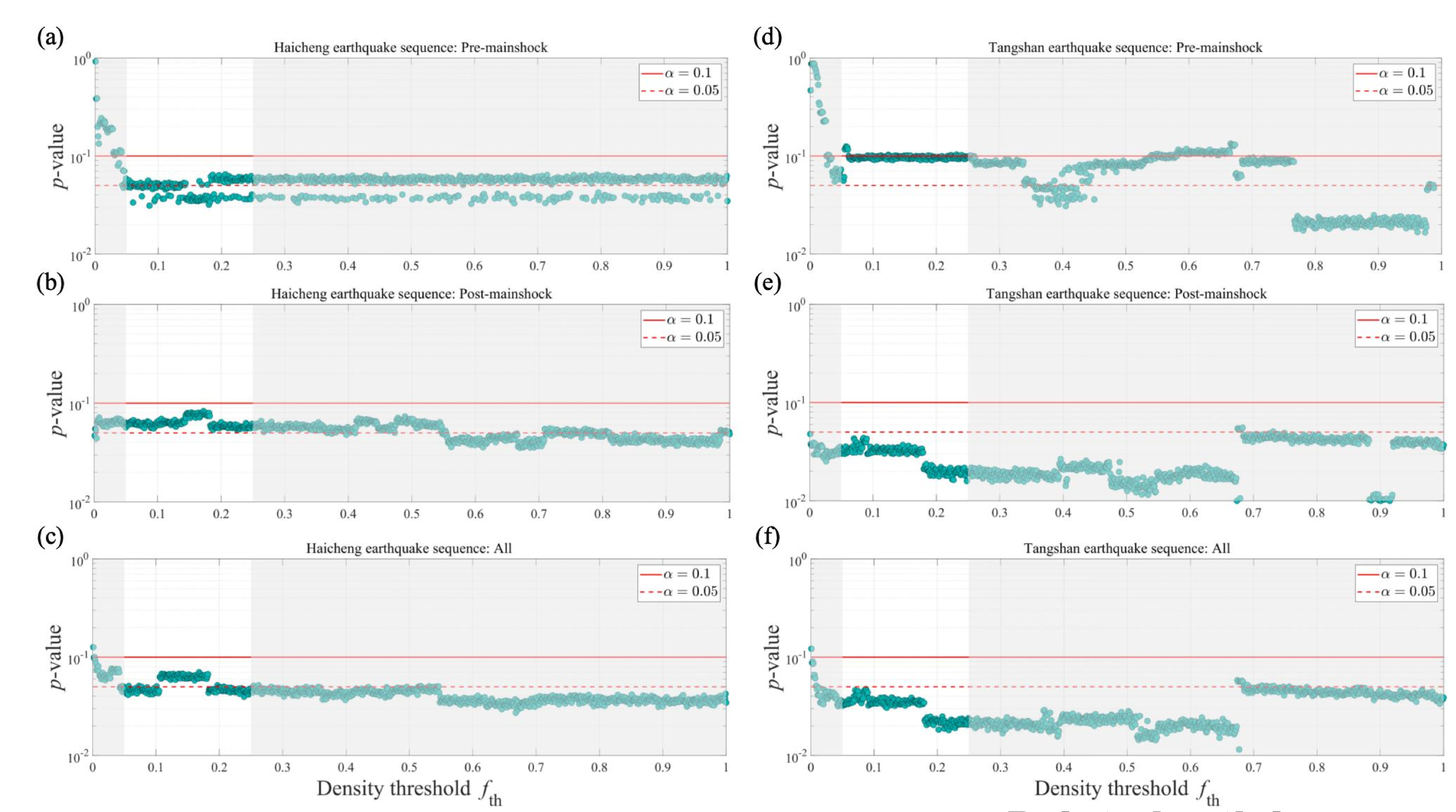

Fig. 4 For the sequences of (a) pre-mainshock, (b) post-mainshock, and (c) combined pre-mainshock and post-mainshock sequences of the m_L 7.4 Haicheng earthquake on February 4, 1975, the p-value of the m_L 7.4 Haicheng earthquake as a potential dragon-king earthquake, obtained with the MRS test statistic using the other events in the sequence, is plotted as function of the density threshold fth. (d), (e), and (f) are same as (a) to (c) for the m_L 7.9 Tangshan earthquake on July 28, 1976. The red solid and dashed lines represent the confidence levels (α = 0.1 and α = 0.05, respectively) commonly used in previous studies. In the white shaded area (for f_{th} approximately between 0.05 and 0.25), we checked that the largest clusters of seismicity are stable and robust.

Key Points:

- Using adaptive kernel density estimation, we propose an objective clustering method that partitions the seismic field into distinct regions.
- We develop a framework to test the dragon-king earthquake hypothesis, using spatial clustering and a high-power outlier detection technique.
- Dragon-king signatures in the Haicheng and Tangshan mainshocks suggest some large and great quakes mature in ways that boost predictability.

Related articles

Li, J., & Sornette, D. (2025). Are Haicheng and Tangshan Earthquakes Dragon-Kings?. (under review) or arXiv preprint arXiv:2504.21310.
Li, J., Sornette, D., Wu, Z., Li, H. (2025). New horizon in the statistical physics of earthquakes: Dragon-king theory and dragon-king earthquakes. Reviews of Geophysics and Planetary Physics, 56(3): 225-242 (in Chinese). doi:10.19975/j.dqyxx.2024-033.

Main references

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de S Cavalcante H L D, Oriá M, Sornette D, et al. 2013. Predictability and control of extreme events in complex systems[J]. Physical Review Letters, 111: 198701.
Sornette D, Ouillon G. 2012. Dragon-kings: Mechanisms, statistical methods and empirical evidence[J]. The European Physical Journal Special Topics, 205(1): 1-26.

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